Systematic and Macroeconomic Long/Short Linear Regression Model

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Regression

1. Objective

Use Multiple Linear Regression Models to forecast 3-month log returns of ETFs (SPY, QQQ, IWM), using macroeconomic and market-based factors.

Future Returns are defined by:

$$y_t = log(P_{t+3}) - log(P_t)$$

Macro Factors

- *Yield Curve = 10y – 2y yields*
- *CPI YoY = inflation trend*
- *Unemployment*
- *Rate Level = 10y treasuries*

Market Signals

- *Momentum (1m, 3m, 12m)*
- *Volatility (6m rolling standard)*
- *Mean Reversion (distance from 12m mean)*

*Note: To reduce asset-specific noise and focus more on the market, the signals are **averaged** across SPY, QQQ, and IWM, providing more aggregate indicators.*

All features are **lagged by one period** before being used in the regression:

$$X_t = features\ at\ (t - 1)$$

This makes sure that predictions are based on available information (minimizing **lookahead bias)**

2. Model Specification

Multiple Linear and Regression Models were used:

- Ordinary Least Squares: Baseline linear model
- Ridge Regression: Minimizes multicollinearity using L2 penalty
- Lasso Regression: Helps with feature importance/selection using L1 penalty
- Elastic Net: Combines effects of L1 and L2 penalties
- Huber Regression: Resistant to outliers using an adaptive residual function
- Quantile Regression (Median): Estimates the median rather than the mean

All of the models follow the general framework of:

$y_t = f(X_t) + \epsilon_t$ where X_t is the macroeconomic and market features

3. Estimation Procedure

I used an expanding window out-of-sample (OOS) approach to help mimic real-world forecasting.

The dataset is split into an initial training period (60 months), after which predictions are generated iteratively:

- At each time t, the model uses the data from 1→ t – 1
- A prediction is made for time t
- The training window expands forward

It helps make sure that all of the predictions are OOS and no future information is used.

4. Evaluation Metrics

The model performance is evaluated using:

- R^2 (coefficient of determination)
- Root Mean Square Error (RMSE): Magnitude of Prediction Errors
- Directional Accuracy
    - $Dir\ Acc\ =\ P(sign(y_t)\ =\ sign(\widehat{y}))$

5. Empirical Results

Across all three ETFs, quantile regression consistently emerges as the best-performing model

ETF	Best Model	R^2	RMSE	Dir Acc
IWM	Quantile	0.041	0.0877	66.9%
QQQ	Quantile	-0.013	0.0775	78.1%
SPY	Quantile	0.032	0.0604	75.8%

5.1 Why Quantile Regression is Performing

While the R^2 values are very close to 0, and negative for some of the alternatives, it is consistent with the fact that asset returns are very unpredictable and that markets are highly efficient. However, there is a strong directional occurrence (despite low explanatory power). The Quantile Regression is able to find the systematic asymmetries purely in terms of return direction.

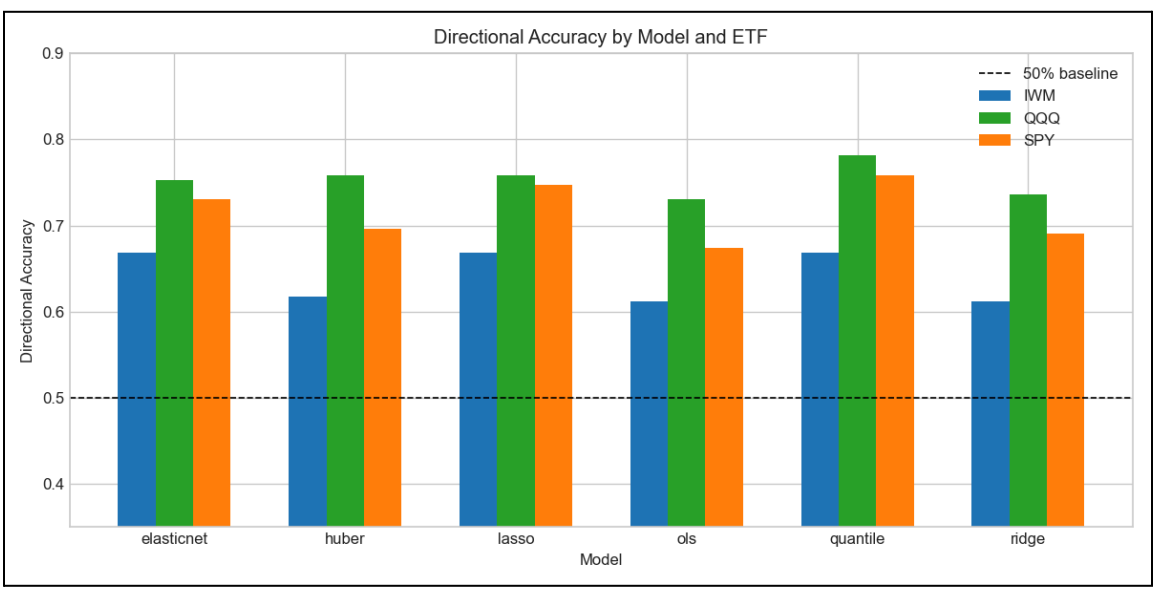
Unlike OLS and other mean-based estimators, quantile regression

- Estimates the conditional median
- Does not assume error distributions
- Is robust to fat tails and outliers

OLS and regularized linear models estimate $E[y_t|X_t]$, while quantile estimates $Median(y_t|X_t)$. In the time period I focused on (2005 – Present), large drawdowns (2008, 2020) disproportionately affect the conditional mean, causing mean-based models to **overreact,** while quantile regression focuses on the median and has more stable predictions.

Additionally, my trading strategy (explained below) focused on directional accuracy. Quantile aligns with it by predicting the central tendency of the conditional distribution and is less influenced by extreme magnitude errors.

While there could/would be multicollinearity between macroeconomic factors that would have benefited from regularization, the more pressing/dominant issue is distributional 'mis-shapings.'



The graph confirms the benefits of the Quantile model. While all models predicted significantly above the baseline (0.5), quantile surpassed all other models for all three ETF's.

Trading Strategy Construction

1. Objective

Translate the regression-based return forecasts into an implementable portfolio strategy. I found that the models show strong directional accuracy (with weaker magnitude prediction), and used that premise to create a strategy surrounding it.

For each ETF, the 'signal' is defined as

$$signal_{i,t} = sign(\widehat{y_{i,t}})$$

- $\widehat{y_{i,t}}$ is the predicted 3-month return from the best-regression model (quantile)

This means that

- +1 → Long position (expected positive return)
- – 1 → Short position (expected negative return)
- 0 → No position

It helps reduce the problem to a directional classification project, helping it align with the strengths of quantile regression.

2. Portfolio Construction

The signals are translated into portfolio weights

$$w_{i,t} = \frac{signal_{i,t}}{\sum_{i}^{n} |signal|_{i,t}}$$

The total exposure becomes 1 as a result, and there is equal capital allocation across active positions.

The normalization is important because (1) it helps prevent excessive leverage, (2) ensures comparability over time, and (3) minimizes concentration risk. Also, since there is no reliable estimate of relative 'alpha', it's better to treat all directions as equally weighted.

I also lagged the portfolio weights by 1 period: $w^{applied} = w_{i,t-1}$ to make sure that all trades at time t are based on information at t – 1

3. Metric Calculations

Gross Returns: $r_t^{gross} = \sum_i w_{i,t} * r_{i,t}$

Turnover: $turnover_t = \sum_i |w_{i,t} - w_{i,t-1}|$

Cost: $cost_t = \ turnover \ * \ 5\ bps \rightarrow r^{gross} = r^{gross} - cost$

These metrics attempt to make the results as realistic as possible, as even modest transaction costs can impact performance. Additionally, I used turnover as a metric to see if the signal is consistent.

4. Backtesting Framework

The backtesting process evaluates the strategy over 177 months (14 years, 9 months), using a fully OOS framework, using the regression outputs

4.1 Performance Metrics

- Annualized Return: 38.69%
- Annualized Volatility: 23.83%
- Sharpe Ratio: 1.44
- Maximum Drawdown: -46.05%
- Hit Rate: 72.32%

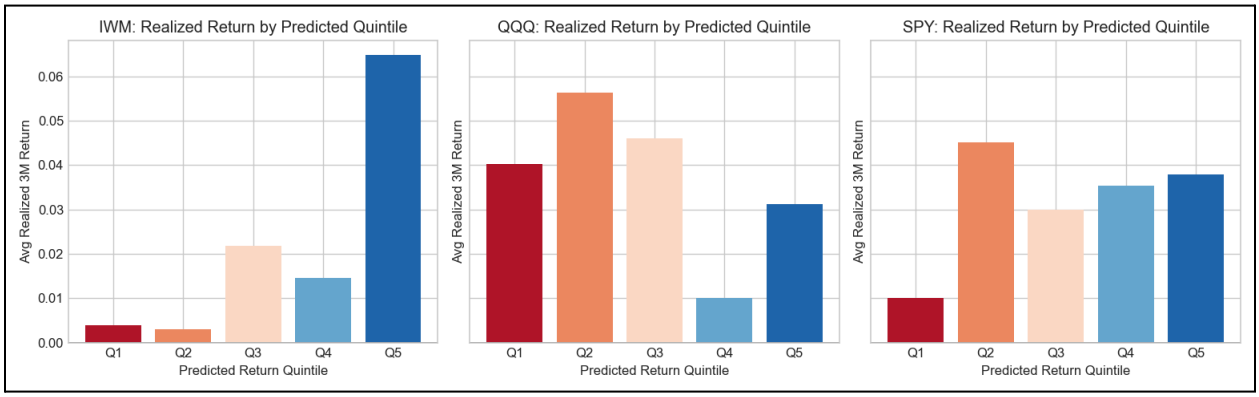
One key note is that we have a significantly high hit rate, despite having a high drawdown as well. It helps corroborate our regression finds, where directional predictability is our primary 'edge'. The high hit rate could imply that despite a majority of the trades being positive, it does lead to smaller gains each time, leading to **more incremental growth**. With the strong performance, we still have systematic market risk exposure during times of economic crises and geopolitical events.

4.2 Portfolio Holdings (Weights)

There is one main pattern: we have extended periods of time of uniform long exposure (⅓ across all ETFs), with the occasional short positions (e.x, IWM during 2022).

This could imply that the model is effectively capturing bull market regimes, where positive returns are prevalent. It is behaving more as a timing model on aggregate equity exposure than a relative value strategy. As with most market-based funds, there is a high correlation between SPY, QQQ, and IWM, possibly showing that the strategy may have some directional beta that contributes to upside and downside.

4.3 Quantile Analysis

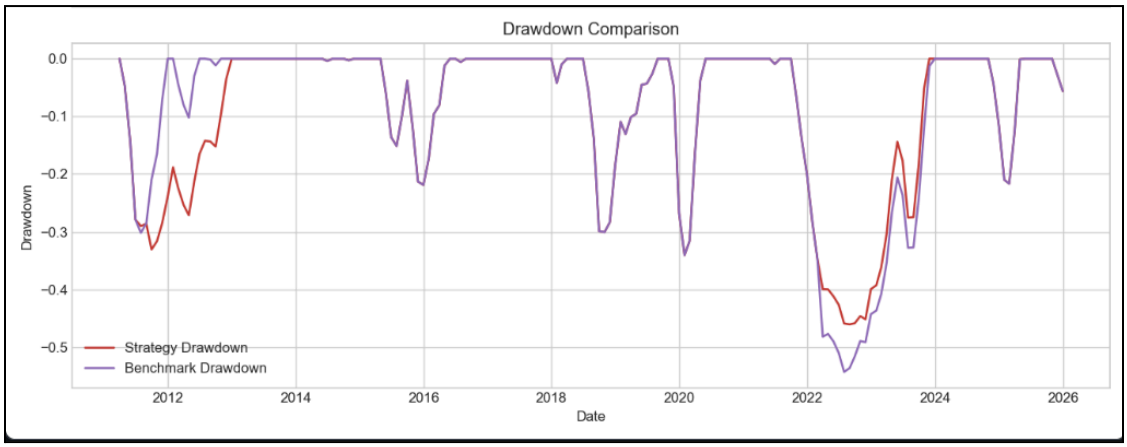


This set of plots groups predictions into quantiles (Q1 – Q5) and evaluates the average return within each predicted bucket. We see strong monotonicity (especially in IWM and SPY) implied that the model successfully ranks the future returns, even though it doesn't precisely estimate them.QQQ shows a less monotonic structure (Q2 and Q3 outperform Q5, with Q4 being relatively weak). The relationship between the predictors and returns may be more complex for tech-heavy equities, or the model may be capturing regime-dependent effects.

This result helps support Quantile Regression and how it outperformed OLS and regularized models:

- OLS minimizes leased squares → focuses on average fit (not ranking)
- Ridge/Lasso shrink coefficients → decrease the useful variation
- **Quantile** Regression helps keep the distributional structure (bigger separation between high and low return regimes).

4.4 Drawdown Analysis



Strategy drawdowns reach almost ~45%, slightly lower in magnitude than the benchmark; however, the strategy oftentimes recovers faster and avoids deeper troughs, helping minimize the overall risk profile.

5. Conclusion

The project set out to evaluate whether regression techniques could be used to construct a systematic trading strategy on ETFs, with a focus on using mean-based models and distribution-focused models. The outperformance of quantile regression comes from the ability to work around the nonlinearities in ETF returns. Financial Markets (and usually in ETFs) show skewness, kurtosis, and regime-dependent behavior.

The overall trading strategy is able to effectively take advantage of our regression output into something that is reasonably stable and profitable. Future iterations will include more technical, idiosyncratic movements in the ETFs to distance ourselves from deriving value mainly from systematic market exposure.